

## 1. Introductory Example:

1.1  $X_i \stackrel{iid}{\sim} \text{Pois}(\lambda)$ , Given  $X_1=1, X_2=2, X_3=4, X_4=2$ .

Estimate  $\lambda$  using MLE, state the estimator, and calculate the estimate.

$$\hat{\lambda} = \bar{X} = \frac{1+2+4+2}{4} = 2.25$$

1.2. Given the same condition, estimate  $P[X=4]$ . State the estimator and calculate the estimate.

$$\hat{P}[X=4] = \frac{\lambda^4 e^{-\lambda}}{4!}, \quad \text{estimate: } \frac{(2.25)^4 e^{-(2.25)}}{4!} = 0.1126$$

1.3 Assume a bag with 3 balls. Each ball is either blue or red.

Let  $X_i$  denote the result of taking one ball from the bag with replacement.

$$X_i = \begin{cases} 1 & \text{if } i\text{th choice is blue} \\ 0 & \text{if } i\text{th choice is red} \end{cases} \quad \text{Assume we have done experiment 4 times, and } X_1=1, X_2=0, X_3=1, X_4=1.$$

That is, first draw is blue, second is red, third is blue, and 4-th draw is blue.

Let number of blue balls in the bag be  $\theta$ , find  $\theta$  using MLE. State the estimator and calculate the estimate.

$$X_i \sim \text{Bern}\left(\frac{\theta}{3}\right) \quad P(X_1=1, X_2=0, X_3=1, X_4=1) = \left(\frac{\theta}{3}\right)\left(1-\frac{\theta}{3}\right)\left(\frac{\theta}{3}\right)\left(\frac{\theta}{3}\right)$$

$$= \left(\frac{\theta}{3}\right)^3 \left(1-\frac{\theta}{3}\right)$$

$\theta$  ranges from 0, 1, 2, 3:

| $\theta$ | 0 | 1    | 2    | 3 |
|----------|---|------|------|---|
| P        | 0 | 2/81 | 8/81 | 0 |

$$\hat{\theta}_{MLE} = 2$$

## 2. Estimator and estimate.

2.1. Assume  $X_i \stackrel{iid}{\sim} \text{Binom}(3, p)$ , find the estimator for  $p$  using MLE.

$$P(X=x|p) = \binom{3}{x} p^x (1-p)^{3-x}, \quad L(\bar{X}_1=x_1, \bar{X}_2=x_2, \dots, \bar{X}_n=x_n|p) = ?$$

$$\binom{3}{x_1} \binom{3}{x_2} \dots \binom{3}{x_n} p^{\sum_{i=1}^n x_i} (1-p)^{\sum_{i=1}^n (3-x_i)} = \text{Likelihood, denote as } L$$

$$\log(L) = \log\left(\binom{3}{x_1}\right) + \log\left(\binom{3}{x_2}\right) + \dots + \log\left(\binom{3}{x_n}\right) + \sum_{i=1}^n x_i \log(p) + \sum_{i=1}^n (3-x_i) \log(1-p)$$

$$\frac{\partial L}{\partial p} = \sum_{i=1}^n x_i \cdot \frac{1}{p} + \sum_{i=1}^n (3-x_i) \cdot \left[-1 \cdot \frac{1}{1-p}\right]$$

$$= \frac{n \cdot \bar{x}}{p} + \frac{n \cdot (3-\bar{x})}{p-1}$$

$$\text{let } \frac{\partial L}{\partial p} = 0; \text{ then}$$

$$\frac{n\bar{x}}{p} = \frac{n(3-\bar{x})}{1-p}$$

$$n\bar{x}(1-p) = n(3-\bar{x})p$$

$$n\bar{x} - n\bar{x}p = 3np - n\bar{x}p$$

$$n\bar{x} = 3np$$

$$\hat{p}_{MLE} = \frac{\bar{x}}{3}$$

2.2. Assume we have observed  $X_1=1, X_2=3, X_3=2, X_4=2$ . What is the estimate for  $p$ ?

$$\hat{p}_{MLE} = \frac{\bar{x}}{3} = \frac{(1+3+2+2)/4}{3} = \frac{2}{3}$$

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2.3 Assume  $X_i \stackrel{iid}{\sim} \text{Exp}(\lambda)$ , find estimator of  $\lambda$  using MLE.

pdf of  $\text{Exp}(\lambda)$  is 
$$\begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$
 let's only consider  $x \geq 0$ .

$$L(\underline{X} | \lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} \quad \log(L) = \sum_{i=1}^n \log(\lambda e^{-\lambda x_i})$$

$$= \sum_{i=1}^n \log(\lambda) + \log(e^{-\lambda x_i})$$

$$= \sum_{i=1}^n \log(\lambda) + (-\lambda x_i)$$

$$\frac{\partial \log L}{\partial \lambda} = \sum_{i=1}^n \frac{1}{\lambda} - x_i$$

$$= \frac{n}{\lambda} - n \cdot \bar{x}$$

let  $\frac{\partial \log L}{\partial \lambda} = 0$  :  $\frac{n}{\lambda} = n \cdot \bar{x}$

$$\hat{\lambda}_{MLE} = \frac{1}{\bar{x}}$$

2.4 Assume we have observed  $X_1 = 1.23$ ,  $X_2 = 3.32$ ,  $X_3 = 1.98$ ,  $X_4 = 2.12$ , what is the estimate for  $\lambda$ ?

$$\hat{\lambda}_{MLE} = \frac{1}{\bar{x}} = \frac{1}{(1.23 + 3.32 + 1.98 + 2.12)/4} = \frac{1}{2.1625} = 0.462$$

3. More general case:

3.1 Now assume  $X_i \stackrel{iid}{\sim} \text{Binom}(m, p)$ . Find the estimator for  $p$  using MLE.

$$P(\bar{X} = x | p) = \binom{m}{x} p^x (1-p)^{m-x} \Rightarrow L(\underline{X} = \underline{x}) = \left( \prod_{i=1}^n \binom{m}{x_i} \right) \cdot p^{\sum_{i=1}^n x_i} \cdot (1-p)^{\sum_{i=1}^n (m-x_i)}$$

$$\log(L) = \sum_{i=1}^n \log\left(\frac{x_i}{m}\right) + \left(\sum_{i=1}^n x_i\right) \log(p) + \left(\sum_{i=1}^n (m-x_i)\right) \log(1-p)$$

$$\frac{\partial \log L}{\partial p} = \left(\sum_{i=1}^n x_i\right) \frac{1}{p} + \left(\sum_{i=1}^n (m-x_i)\right) \cdot \frac{1}{p-1}$$

$$= \frac{n \cdot \bar{x}}{p} + \frac{n(m-\bar{x})}{p-1}$$

$$\text{let } \frac{\partial \log L}{\partial p} = 0$$

$$\frac{n \bar{x}}{p} = \frac{n(m-\bar{x})}{1-p}$$

$$n \bar{x}(1-p) = n(m-\bar{x})p$$

$$n \bar{x} - n \bar{x} p = n m p - n \bar{x} p$$

$$\hat{p}_{MLE} = \frac{\bar{x}}{m}$$

#### 4. Real-Life Problem.

4.1. Randomly sample 100 mosquitoes, 44 of them carry parasites.

Derive the proportion of infected mosquitoes in the population using MLE.

denote the proportion as  $p$ , then the number of infected mosquitoes in sample follows binomial distribution,  $\text{Binom}(m, p)$ . In this case,  $m=100$ ,  $x=44$

$$L(X=44|P) = \binom{100}{44} p^{44} (1-p)^{56}$$

$$\log L = \log\left(\binom{100}{44}\right) + \log(p^{44}) + \log((1-p)^{56})$$

$$= \log\left(\binom{100}{44}\right) + 44 \log p + 56 \log(1-p)$$

$$\frac{\partial \log L}{\partial p} = 44 \cdot \frac{1}{p} + 56 \cdot \frac{1}{p-1} \quad \text{let } \frac{\partial \log L}{\partial p} = 0 :$$

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$$\frac{44}{p} = \frac{56}{p-1} \Rightarrow 44 - 44p = 56p$$

So our estimate is  $\hat{p}_{MLE} = 0.44$

$$100p = 44$$

$$\hat{p}_{MLE} = \frac{44}{100} = 0.44$$

4.2 What is the estimator for  $N$ , total number of tanks, in the German Tank problem?

Assume we captured  $k$  tanks. let  $X$  denote the # of tanks we can capture

$$P(X=k | N) = \frac{1}{\binom{N}{k}} \quad \text{since there are } \binom{N}{k} \text{ ways of capturing.}$$

$$\arg \max_N P(X=k | N) = ?$$

Notice that

$$\frac{1}{\binom{N}{k}} \rightarrow 0 \quad \text{as } N \rightarrow \infty$$

So we want  $N$  to be small as possible,

$$\text{But } N \geq \max\{X_1, X_2, \dots, X_k\}. \quad \text{So } \hat{N}_{MLE} = \max\{X_1, \dots, X_k\}.$$

3.2 Assume  $\bar{X}$ : iid Geom( $p$ ), Find estimator for  $p$  using MLE.

$$P(X=x|p) = (1-p)^{x-1} p.$$

$$L(\bar{X}=X|p) = \prod_{i=1}^n (1-p)^{x_i-1} p = (1-p)^{\sum_{i=1}^n (x_i-1)} p^n$$

$$\log(L) = [\sum_{i=1}^n (x_i-1)] \log(1-p) + n \log p.$$

$$\frac{\partial \log L}{\partial p} = [\sum_{i=1}^n (x_i-1)] \frac{1}{p-1} + \frac{n}{p}.$$

$$\text{Setting } \frac{\partial \log L}{\partial p} = 0;$$

$$\frac{\sum_{i=1}^n (x_i-1)}{1-p} = \frac{n}{p} \Rightarrow n - np = n\bar{x}p - np$$

$$n = n\bar{x}p$$

$$\hat{p}_{MLE} = \frac{1}{\bar{x}}.$$

Does it look familiar?